
Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

Anonymous Author(s)

Affiliation

Address

email

Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 22 tissue sections from four technologies (336 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, implicit neural-field and imputation baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, so we assembled 10 independent specimens across five organs and two donors’ species. At that level the advantage survives family-wise correction against 3 baselines (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN); Holm-corrected $p \leq 0.020$, winning on at least 9/10 specimens), including one graph model. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r , more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.002, against 0.014 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the strongest graph baseline the separation is still unresolved (STAGATE-style: 7/10 specimens, $p = 0.43$), so the method is distinguishable from most of this literature but not from all of it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.

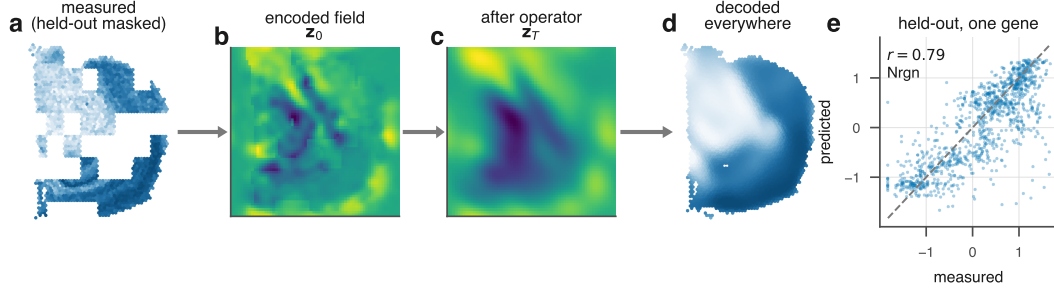


Figure 1: **Neural Reaction-Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction-diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction-diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction-Diffusion Operators (NRDO)**, which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction-diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction-diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions ??–1); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts our perturbation result.

66 2 Method

67 A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized
68 to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode–
69 evolve–decode.

70 **Spatial encoder.** Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al.,
71 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement
72 to a per-channel gain, so the layer depends on geometry rather than node identity. Features are
73 splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and
74 an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral
75 blocks [Li et al., 2021] refine the field.

76 **The reaction–diffusion operator.** The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

77 with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so
78 definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as
79 1×1 convolutions with a tanh-bounded output.

80 Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t =$
81 $O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the
82 Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

83 Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-
84 step map that is second-order accurate (Proposition ??). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2)
85 lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean
86 exactly, with no CFL restriction (Proposition ??). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this
87 holds along the entire optimization trajectory — no step size couples to learned parameters, and no
88 projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-
89 mean component (Proposition ??). Both are proved in Appendix ?? and checked numerically in
90 Table ??; Section ?? measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths
91 per unit pseudo-time, convertible to microns via the stored coordinate scale.

92 **Decoder.** Predictions come from bilinear interpolation of the evolved field at any continuous co-
93 ordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could
94 memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation
95 uninterpretable.

96 **Objectives.** The objective combines reconstruction (MSE plus a per-gene Pearson term, since the
97 spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and
98 physics terms. The conventional physics-informed penalty—the PDE residual along the model’s
99 own trajectory—turns out to be uninformative for an exactly integrated operator.

100 Proposition 1 makes this precise: the residual along the model’s own trajectory is $\frac{\Delta t}{2} \|[\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta] \mathbf{z}\| +$
101 $O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing
102 it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward
103 operators whose diffusive and reactive parts commute (Appendix A). We instead require that the
104 *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

105 the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$
106 and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

107 **Interpretability.** Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_k$ with
108 $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned
109 operator — a falsifiable read-out of the dynamics it has inferred (Appendix ??).

110 3 Experimental setup

111 **Data.** We use 12 public datasets spanning four spatial technologies (23 tissue sections, 984,112
112 measured locations; Fig. ??), plus one non-spatial perturbation screen (111,659 dissociated cells)
113 used only in Section ?. Raw reads are never used: we consume vendor count matrices and record a
114 SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization,
115 log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so
116 the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates
117 are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned
118 diffusion coefficient meaningless.

119 **Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the en-
120 coder and every model predicts expression at those coordinates. Random spot-level splits leak, since
121 a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous
122 blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

123 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
124 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
125 et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit
126 neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods
127 were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance
128 read-out head — the most favorable choice available — and is marked *-style*, since these are not
129 reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and
130 evaluation code.

131 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,
132 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because
133 correlation and RMSE both reward regression toward a smooth mean.

134 **Two scopes, kept distinct.** Results are reported at two scales and every number is labeled with
135 the one it belongs to. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and is the
136 basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full
137 budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher
138 on the primary section, and a value from one is never quoted in support of a claim about the other.

139 4 Results

140 4.1 Masked spatial reconstruction across 22 sections

141 NRDO improves masked-region reconstruction over every baseline tested, on 22 sections spanning
142 four technologies (336 runs), and the improvement survives the conservative unit of analysis (Ta-
143 ble 1).

144 Two units are reported because they answer different questions. With the *section* as the unit every
145 model sees identical held-out blocks, so a paired test removes the between-section variance that
146 otherwise dominates; all 7 baselines are beaten with Holm-corrected $p \leq 0.003$. But twelve of these
147 sections are serial sections of one brain, and treating them as independent overstates the effective
148 sample size. Collapsing to 10 independent *specimens* costs most of that power and is the defensible
149 test: the separation still survives family-wise correction against 3 baselines (GP, multi-bandwidth,

Table 1: **Masked spatial reconstruction across 22 sections and 10 independent specimens.** Left: mean $\pm$ s.d. over sections, with Holm-corrected stars from the paired section-level test. Right: the conservative analysis, in which serial sections of one specimen are collapsed before testing — specimens won, Cohen’s d_z , and the Holm-corrected p over the baseline family, bold where $p < 0.05$. $|\Delta I|$ is the error in Moran’s I , on which NRDO is worst: the predicted fields are too smooth. Section-level $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

model	Pearson $r \uparrow$	$ \Delta I \downarrow$	specimens	d_z	p_{Holm}
STAGATE-style	$0.170 \pm 0.064^{**}$	0.769 ± 0.045	7/10	0.45	0.43
Autoencoder (non-spatial)	$0.169 \pm 0.063^{***}$	0.761 ± 0.049	7/10	0.65	0.21
GNN (GCN)	$0.167 \pm 0.061^{***}$	0.793 ± 0.046	9/10	1.05	0.020
Tangram-style	$0.163 \pm 0.064^{***}$	0.805 ± 0.054	9/10	0.92	0.08
GP, multi-bandwidth	$0.133 \pm 0.053^{***}$	0.770 ± 0.040	10/10	2.12	0.014
Neural field (SIREN)	$0.118 \pm 0.051^{***}$	0.612 ± 0.064	10/10	1.93	0.014
NRDO (ours)	0.183 ± 0.063	0.848 ± 0.047	—	—	—

Neural field (SIREN), GNN (GCN)), winning on at least 9/10 specimens at $p \leq 0.020$. It does not survive against STAGATE-style, which we return to in Section 6.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.169 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to 0.002 Pearson r , against 0.014 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms (Section 5).

Spatial autocorrelation. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.335 against 0.347), and it has by some margin the largest error in Moran’s I (0.848 against 0.612 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

5 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (−0.023), and removing the reaction gives 0.241 ± 0.003 (−0.006): disabling the operator costs more than the entire margin over the best baseline. The control is exactly parameter-matched — the operator holds 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused — so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the control separates the two explanations. The full sweep, including the biological regularizers and the PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in Appendix ??.

6 Limitations

Identifiability. A single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport; Section 4 shows they are moreover close to their initialization, which is why we make no claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

Over-smoothing. The predicted fields are smoother than the measurements, and on Moran’s I error NRDO trails the baselines it leads on the other metrics. This survives training to convergence and is only partly correctable by a spectral-matching term (Section ??), so it bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so.

One baseline remains unresolved. The benchmark’s statistical resolution is set by independent specimens, not sections: twelve of 22 sections are serial sections of one brain. An earlier version of this work reported five specimens, at which no family-wise significant result was reachable at any effect size. We therefore added tissues rather than sections, reaching 10, and the separation now survives Holm correction against 3 baselines including GNN (GCN) ($p = 0.020$, 9/10 specimens). It does not against STAGATE-style (7/10, $p = 0.43$), whose attention-weighted neighbourhood aggregation is the closest existing analogue to what the operator does. We take that as the honest boundary of the claim: the continuous formulation is distinguishable from most of this literature and not from all of it. One section (visium_human_heart) is excluded from every specimen-level analysis because its held-out set falls below 200 locations, at which Pearson r cannot be estimated to the precision the comparison requires.

Transfer. Under the held-out-block protocol the fitted operator does not transfer to a new tissue at all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline transfers better. It is a within-section model, and we no longer claim otherwise.

Scope. The *-style* baselines implement each method’s architectural core with a common read-out head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget and subsampled sections so that 336 runs were feasible on CPU, and scope is limited to single 2-D sections.

References

- Benjamin Paul Chamberlain, James Rowbottom, Maria Gorinova, Stefan Webb, Emanuele Rossi, and Michael M. Bronstein. GRAND: Graph neural diffusion. In *International Conference on Machine Learning (ICML)*, 2021.
- Kok Hao Chen, Alistair N. Boettiger, Jeffrey R. Moffitt, Siyuan Wang, and Xiaowei Zhuang. Spatially resolved, highly multiplexed RNA profiling in single cells. *Science*, 348(6233):aaa6090, 2015.
- Jeongwhan Choi, Seoyoung Hong, Noseong Park, and Sung-Bae Cho. GREAD: Graph neural reaction diffusion networks. In *International Conference on Machine Learning (ICML)*, 2023.
- Kangning Dong and Shihua Zhang. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nature Communications*, 13(1):1739, 2022.
- Alfred Gierer and Hans Meinhardt. A theory of biological pattern formation. *Kybernetik*, 12(1):30–39, 1972.
- Jian Hu, Xiangjie Li, Kyle Coleman, Amelia Schroeder, Nan Ma, David J. Irwin, Edward B. Lee, Russell T. Shinohara, and Mingyao Li. SpaGCN: Integrating gene expression, spatial location and

227 histology to identify spatial domains and spatially variable genes by graph convolutional network.
228 *Nature Methods*, 18(11):1342–1351, 2021.

229 Anna Kicheva, Periklis Pantazis, Tobias Bollenbach, Yannis Kalaidzidis, Thomas Bittig, Frank
230 Jülicher, and Marcos González-Gaitán. Kinetics of morphogen gradient formation. *Science*, 315
231 (5811):521–525, 2007.

232 Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional net-
233 works. In *International Conference on Learning Representations (ICLR)*, 2017.

234 Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, An-
235 drew Stuart, and Anima Anandkumar. Neural operator: Graph kernel network for partial differen-
236 tial equations. *arXiv preprint arXiv:2003.03485*, 2020.

237 Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, An-
238 drew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential
239 equations. In *International Conference on Learning Representations (ICLR)*, 2021.

240 Patrick A. P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950.

241 Patrick Müller, Katherine W. Rogers, Ben M. Jordan, Joon S. Lee, Drew Robson, Sharad Ra-
242 manathan, and Alexander F. Schier. Differential diffusivity of Nodal and Lefty underlies a
243 reaction-diffusion patterning system. *Science*, 336(6082):721–724, 2012.

244 Giovanni Palla, Hannah Spitzer, Michal Klein, David Fischer, Anna Christina Schaar,
245 Louis Benedikt Kuemmerle, Sergei Rybakov, et al. Squidpy: A scalable framework for spatial
246 omics analysis. *Nature Methods*, 19(2):171–178, 2022.

247 Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*.
248 MIT Press, 2006.

249 David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-
250 Arroita, Severin Hauenstein, et al. Cross-validation strategies for data with temporal, spatial,
251 hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

252 Rushikesh Sheth, Luciano Marcon, M. Félix Bastida, Marisa Junco, Laura Quintana, Randall Dahn,
253 Marie Kmita, James Sharpe, and Maria A. Ros. Hox genes regulate digit patterning by controlling
254 the wavelength of a Turing-type mechanism. *Science*, 338(6113):1476–1480, 2012.

255 Patrik L. Ståhl, Fredrik Salmén, Sanja Vickovic, Anna Lundmark, José Fernández Navarro, Jens
256 Magnusson, Stefania Giacomello, et al. Visualization and analysis of gene expression in tissue
257 sections by spatial transcriptomics. *Science*, 353(6294):78–82, 2016.

258 Gilbert Strang. On the construction and comparison of difference schemes. *SIAM Journal on Nu-*
259 *merical Analysis*, 5(3):506–517, 1968.

260 Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan,
261 Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let
262 networks learn high frequency functions in low dimensional domains. In *Advances in Neural*
263 *Information Processing Systems (NeurIPS)*, 2020.

264 Alan M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal*
265 *Society of London B*, 237(641):37–72, 1952.

266 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez,
267 Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Infor-*
268 *mation Processing Systems (NeurIPS)*, 2017.

- 269 Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment:
270 From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–
271 612, 2004.
- 272 Lewis Wolpert. Positional information and the spatial pattern of cellular differentiation. *Journal of*
273 *Theoretical Biology*, 25(1):1–47, 1969.
- 274 Shuizi Rachel Yu, Markus Burkhardt, Matthias Nowak, Jonas Ries, Zdeněk Petrášek, Steffen
275 Scholpp, Petra Schwille, and Michael Brand. Fgf8 morphogen gradient forms by a source-sink
276 mechanism with freely diffusing molecules. *Nature*, 461(7263):533–536, 2009.

277 A Vacuity of the along-trajectory residual

278 Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own
279 trajectory. The following makes precise why that penalty is uninformative for the present architec-
280 ture, and motivates the steady-state formulation adopted instead.

281 **Proposition 1** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$
282 $\Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-
283 ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (4)$$

284 where $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,
285 and $R_{\Delta t}$ does not depend on the observed data.

286 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
287 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
288 evaluated at the same state $\mathbf{z}$. $\square$

289 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
290 observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts
291 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
292 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
293 scaling is confirmed in Table ?? (observed order 1.00).

294 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
295 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
296 constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space,
297 characterized next.

298 B Non-identifiability from a single snapshot

299 The following records the central epistemic limitation of the setting: an operator fitted to one station-
300 ary field is determined only on a measure-zero subset of its own domain. The argument is elementary
301 — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint
302 simply says nothing off that image — and we state it as a theorem for its consequence rather than
303 for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in
304 this literature: it means every counterfactual an operator of this class produces is a statement about
305 the architecture, not about the data, and it predicts in advance the null result of Section ??.

306 **Theorem 2** (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose
307 $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (5)$$

308 *If $\mathcal{F}(\mathbf{D}, \mathbf{z}^*) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for
309 every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the
310 space of such φ is infinite-dimensional.*

311 *Proof.* The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it
312 whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$
313 is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets
314 of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$
315 is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ;
316 choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a
317 linearly independent family, all vanishing on $\mathcal{R}$. $\square$

318 **Corollary 3** (What a stationarity fit can determine). *Under the hypotheses of Theorem 2, the steady-*
319 *state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about*
320 *f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the*
321 *observed manifold—is determined by the inductive bias of the architecture and the regularizers, not*
322 *by the data.*

323 Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section ?? dis-
324 places $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states
325 off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is
326 therefore consistent with the theory rather than merely an empirical disappointment: a single sta-
327 tionary snapshot does not contain the information such a counterfactual would require. Breaking the
328 degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal
329 sampling, or interventional data.